

and cutting the polygon into quadrilateral regions. After Baryshnikov, one can define a compatibility relation between quadrangulations. On the set R_Q of quadrangulations compatible with Q , there is a mutation process, that defines a mutation graph. Moreover, this mutation process is oriented and defines a partial order. The mutation graph is also the skeleton of a polytope, called the Stokes polytope for the quadrangulation [MP19, Man18, BMP17].

There is a simple way to map the planar-embedded arbor t into a quadrangulation. There is also a way to define a gentle quiver from any quadrangulation. The composite map recovers the definition of G_t from t . By this correspondence, the τ -tilting theory exactly matches the combinatorial description using mutations of compatible quadrangulations.

It follows that the conjectural statements above could be rephrased entirely using the structures (mutation graph, poset, polytope) coming from this context instead of those coming from τ -tilting theory.

6 Arbors of type $\mathbb{A}$

In this section, we consider the case of linear arbors in which each vertex has one element and at most one sub-tree, so that their shape is a simple chain. This is the intersection of the set of unary-binary arbors considered before with the set of linear arbors. In order to compute the Zeta polynomial and the M -triangle for the poset P_t in this case, it is convenient to study a more general family of posets.

In this section, we will use the following convention: for a product $\beta_\ell = \prod_{r=2}^\ell \alpha_r$, the value at $\ell = 1$ is 1 and the value at $\ell = 0$ is $1/\alpha_1$.

Let $m \geq 1$ be an integer, used as a slope parameter. Let R_m be the region in the plane of coordinates (x, y) defined by the conditions

$$0 \leq x, \quad 0 \leq y, \quad my < x. \quad (28)$$

Let now (x, y) be nonnegative integers such that

$$my < x. \quad (29)$$

We consider the set of lattice paths in R_m that go from $(0, 0)$ to the fixed point (x, y) , by steps $(+1, 0)$ (denoted by X) or $(0, +1)$ (denoted by Y). The point $(0, 0)$ is the unique allowed point on the line $my = x$ at the boundary of R_m .

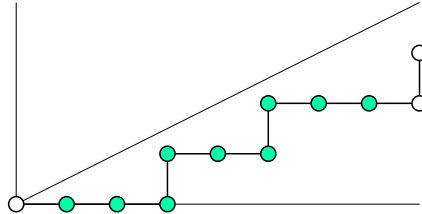


Figure 2: Example of lattice path $X^3YX^2YX^3Y$ with $m = 2$, $x = 8$ and $y = 3$. The word A associated with vertical steps is $(0, 0, 1, 0, 1, 0, 0)$.